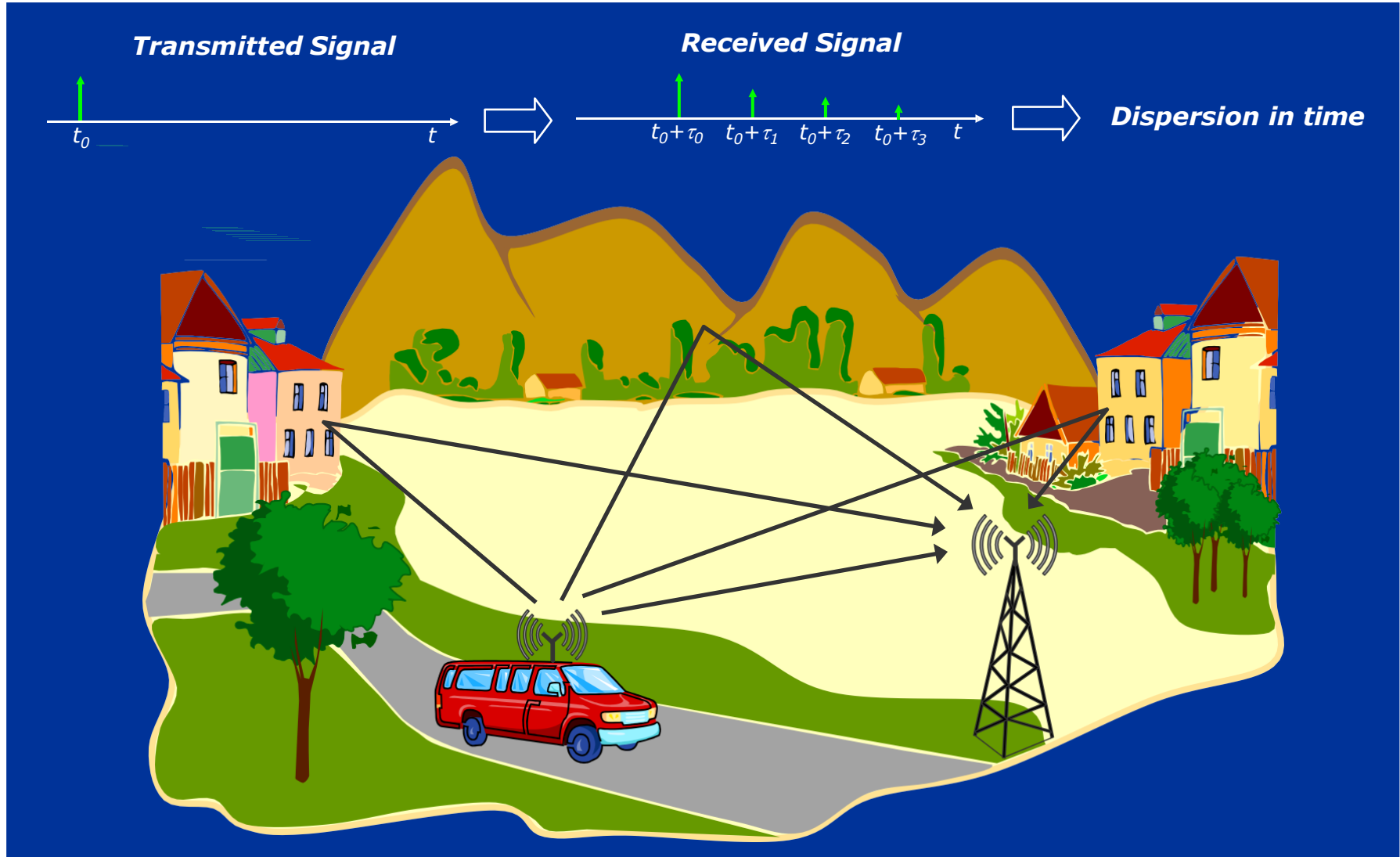
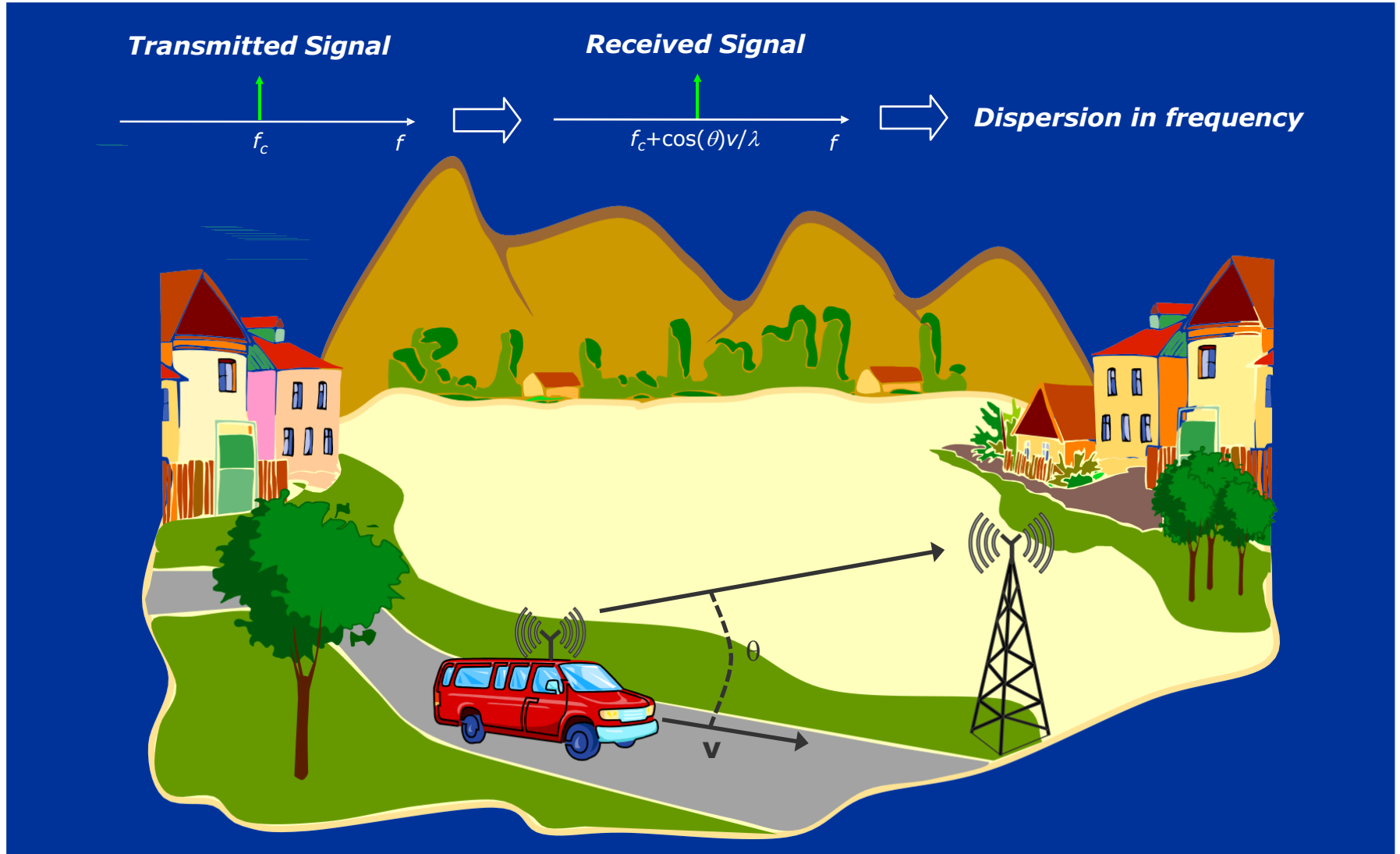


To study and perform
the importance of cyclic
prefix in OFDM using
IDFT and DFT blocks.

The Wireless Propagation Channel

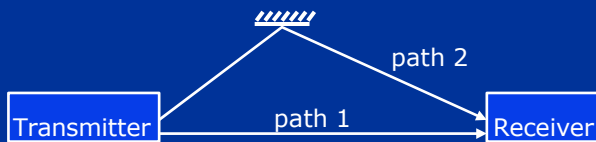


The Wireless Propagation Channel



The Wireless Propagation Channel

Two-path channel at time t_1



Two-path channel
relative delay = T'

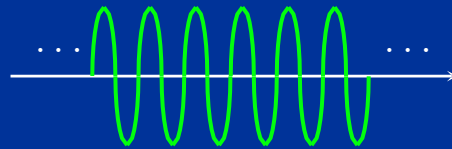
Two-path channel at time t_2



Two-path channel
relative delay = $T'' = T'/2$

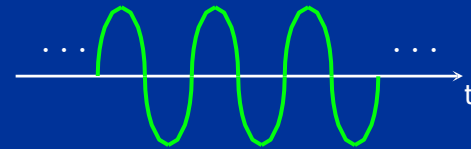
Transmitted Signal 1

$$f_1 = 1/T' = 1/(2T'')$$

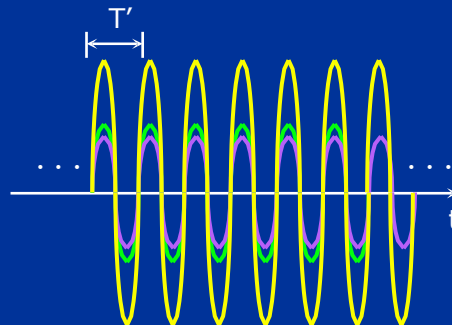


Transmitted Signal 2

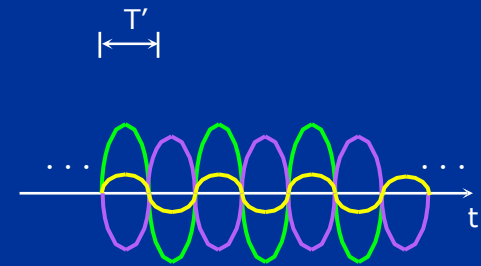
$$f_2 = 1/(2T') = 1/(4T'')$$



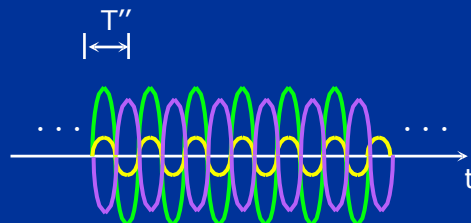
Received Signal 1 at t_1



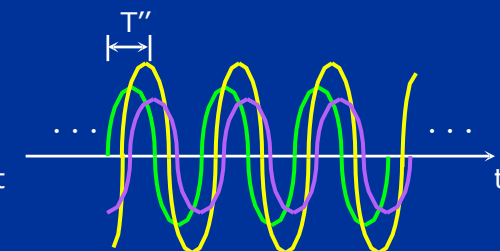
Received Signal 2 at t_1



Received Signal 1 at t_2



Received Signal 2 at t_2

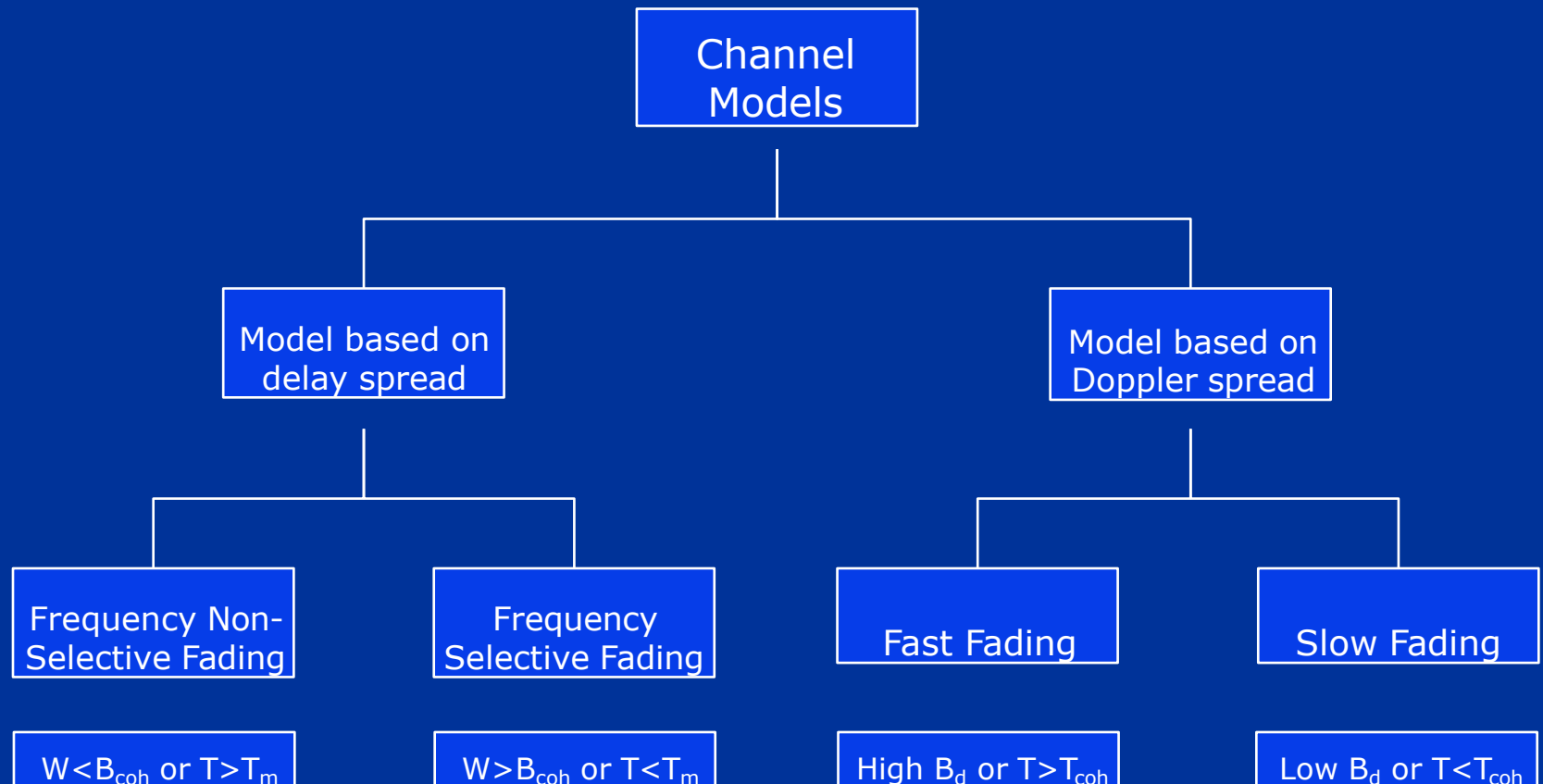


Time Selective Fading

Frequency Selective Fading

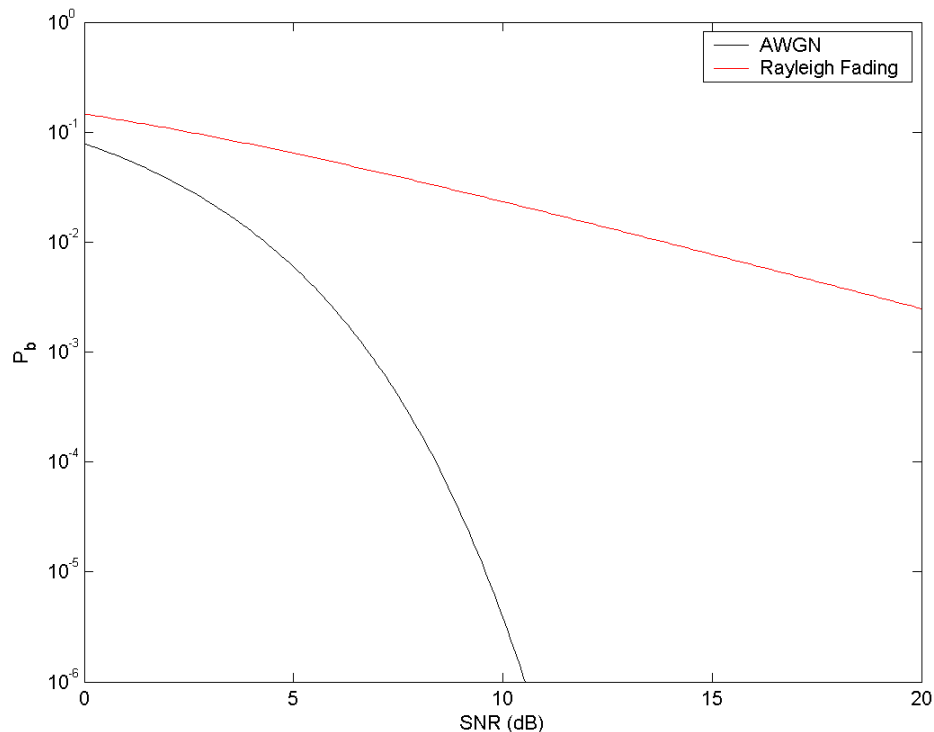
Channel Models: Classification *and Characteristics*

Channel Models Classification and Characteristics



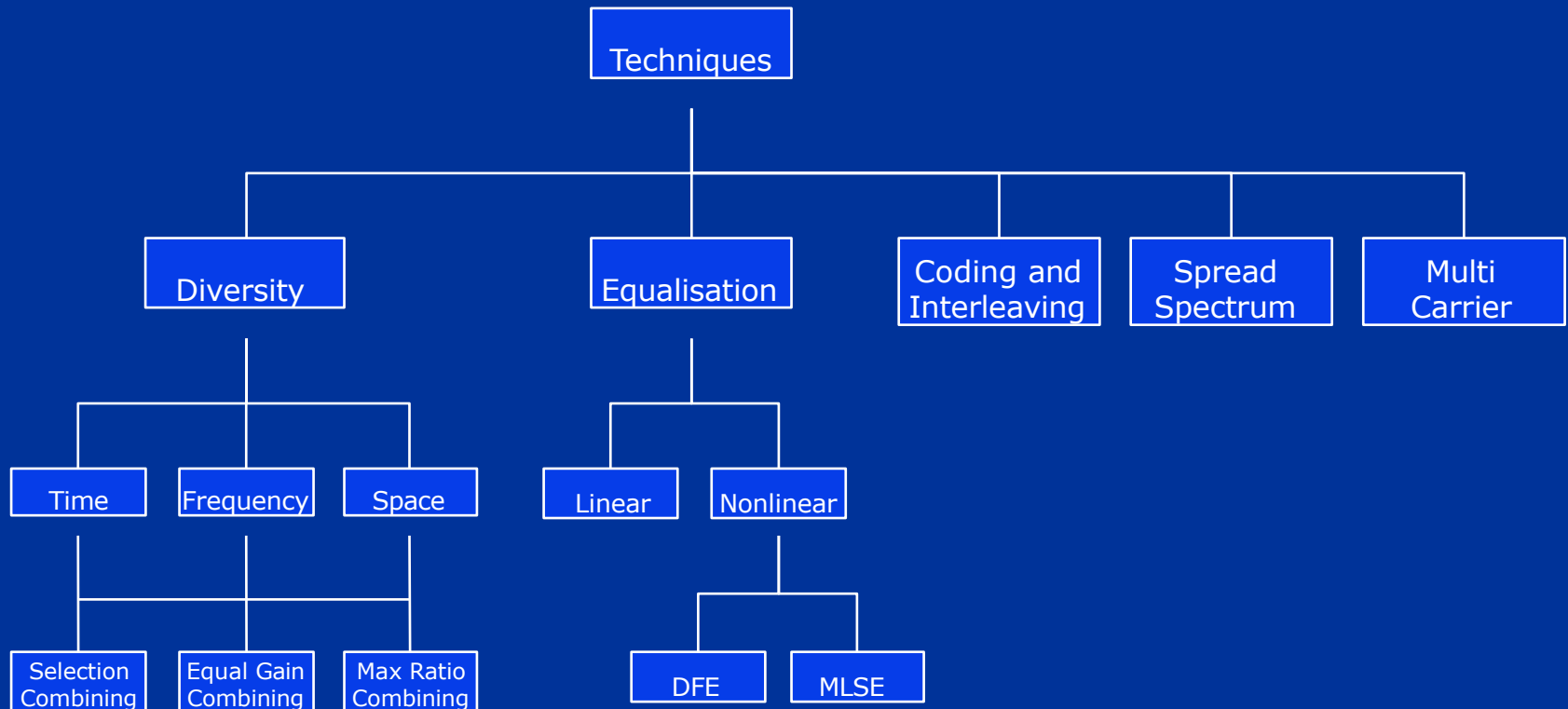
The Frequency Non-Selective, Slow Rayleigh Fading Channel

P_b vs. SNR for a BPSK signal in an AWGN channel and in a Rayleigh fading channel



Techniques to Overcome Time *and Frequency Selectivity*

Techniques to Overcome Time and/or Frequency Selectivity



Orthogonal Frequency Division *Multiplexing*

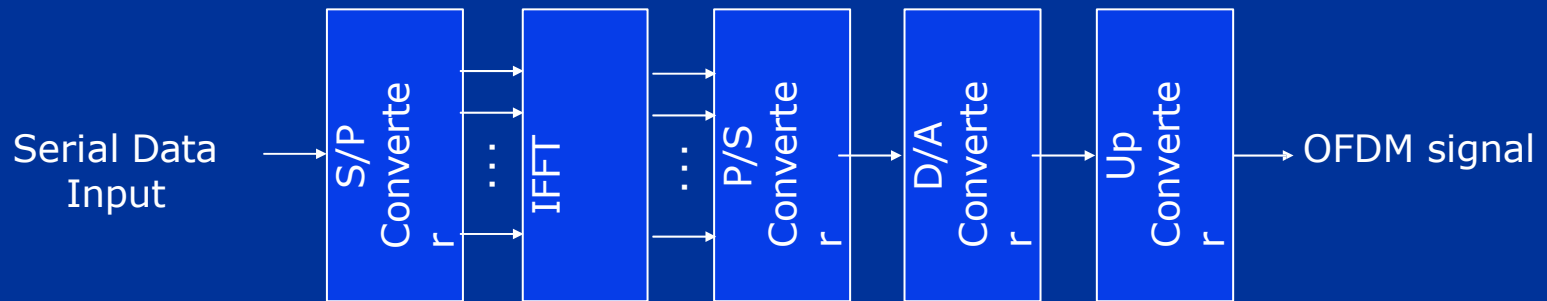
- OFDM is a multi-carrier modulation scheme.
- In OFDM the frequency spacing between adjacent sub-carriers is $\Delta f = 1/T_{MC} = 1/(NT_{SC})$.
- $\Delta f = 1/T_{MC}$ is the minimum frequency separation that is necessary to ensure orthogonality between the sub-carriers over the signalling interval of length T_{MC} .
- In OFDM the frequency spectrum of each sub-carrier overlaps the frequency spectrum of adjacent sub-carriers.

The idea behind using OFDM is to combat **frequency selective fading**, where different frequency components of a transmitted signal can undergo different levels of fading.

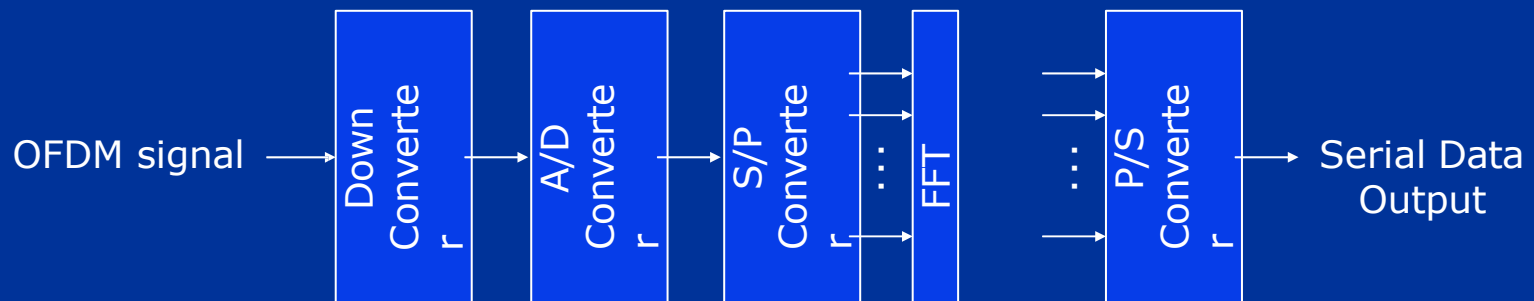
The OFDM divides the available spectrum into small chunks of subchannels. Within the subchannels, the fading experienced by the individual modulated symbol can be considered flat.

Orthogonal Frequency Division *Multiplexing*

OFDM Transmitter



OFDM Receiver



Orthogonal Frequency Division *Multiplexing*

■ Advantages of OFDM:

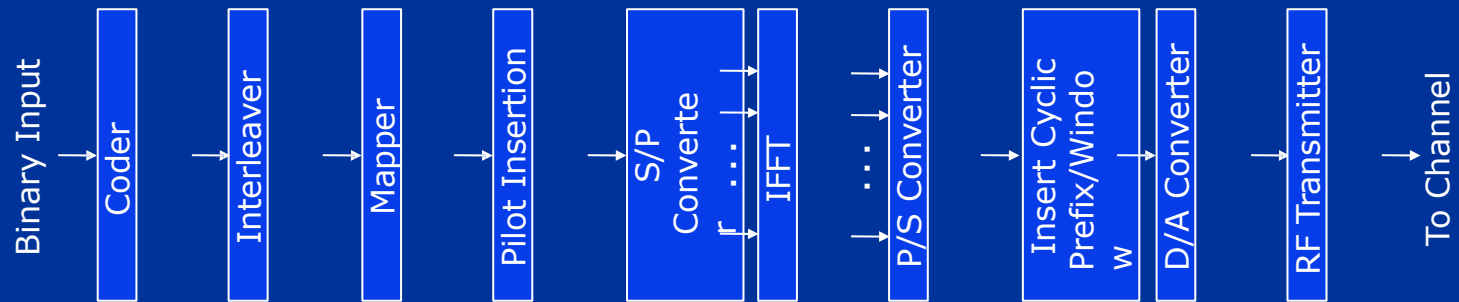
- Good performance under delay spread/frequency selective fading conditions;
- Bandwidth efficiency;
- Efficient digital signal processor based generation/detection techniques.

■ Disadvantages of OFDM:

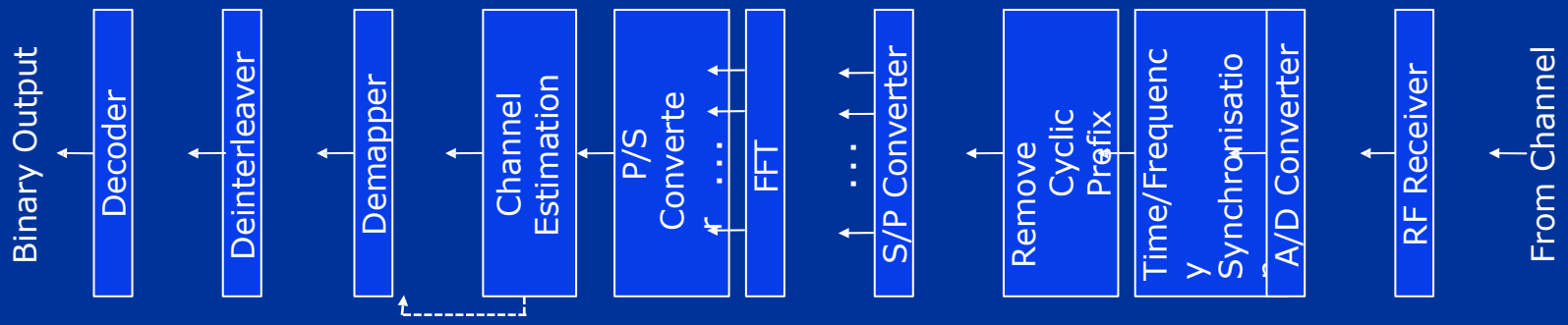
- Poor performance under Doppler spread/time selective fading conditions;
- Sensitive to non-linear distortion;
- Sensitive to timing and frequency offsets as well as phase noise.

Elements of a Typical OFDM Communications System

OFDM Transmitter



OFDM Receiver

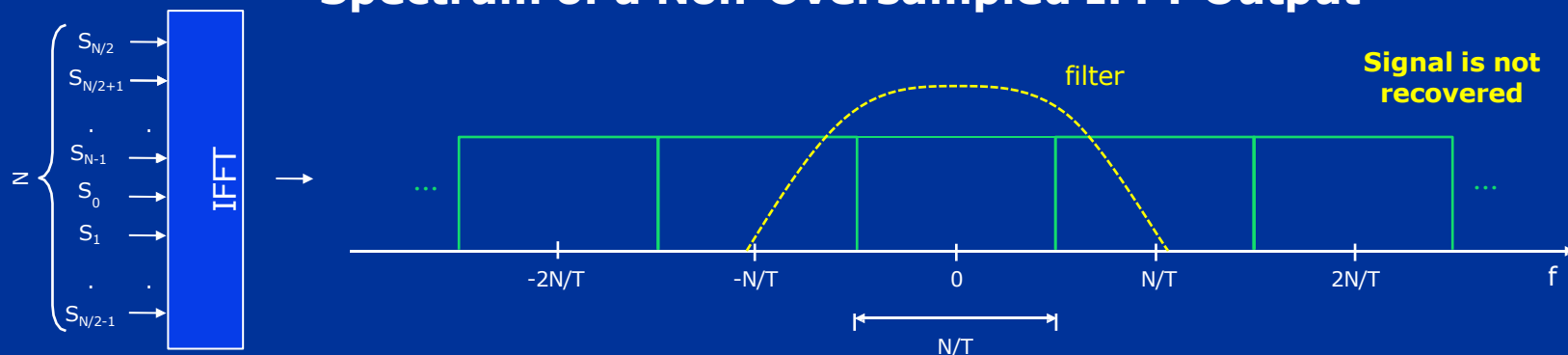


Zero Padding

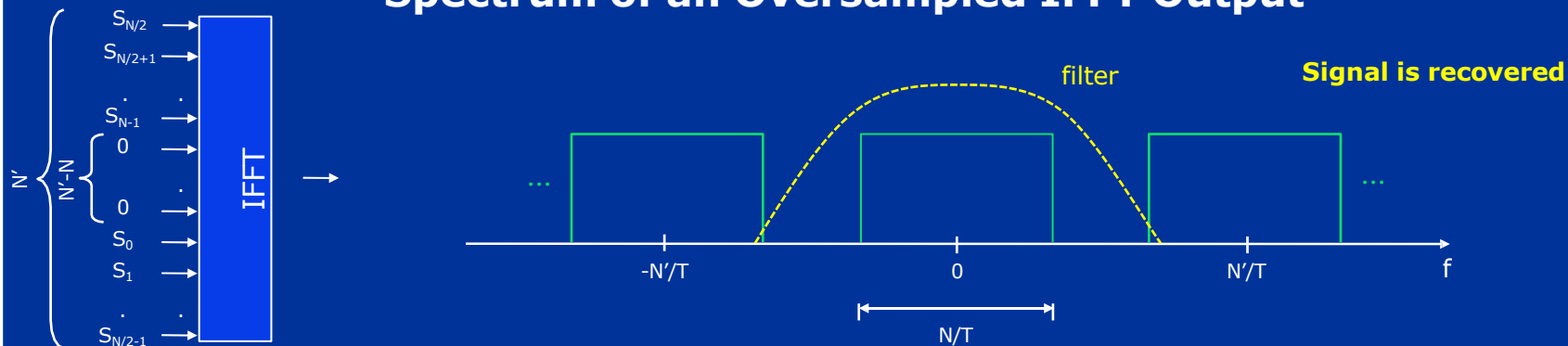
- With an N point IDFT, it is only possible to generate an N sub-channel OFDM symbol sampled at the Nyquist rate.
- In this case, it is difficult to recover the continuous time signal from the sampled signal with filters with realisable passband-to-stopband transition regions.
- With an $N' > N$ point IDFT, it is possible to generate an N sub-channel OFDM symbol sampled at a rate higher than the Nyquist rate.
- In this case, it is easier to recover the continuous time signal from the oversampled signal using filters with realisable passband-to-stopband transition regions.
- The zero padding technique achieves oversampling.

Zero Padding

Spectrum of a Non-Oversampled IFFT Output



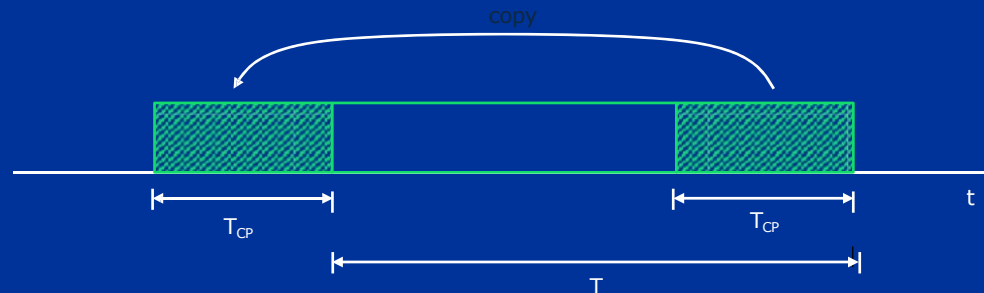
Spectrum of an Oversampled IFFT Output



Cyclic Prefix

- To eliminate ISI in OFDM a guard time is inserted with a duration longer than the multipath channel maximum delay.
- Moreover, to eliminate ICI in OFDM the guard time is cyclically extended.
- Note that in a multipath channel an appropriate guard time avoids ISI but not ICI, unless it is cyclically extended.

Cyclic Extension of an OFDM Symbol



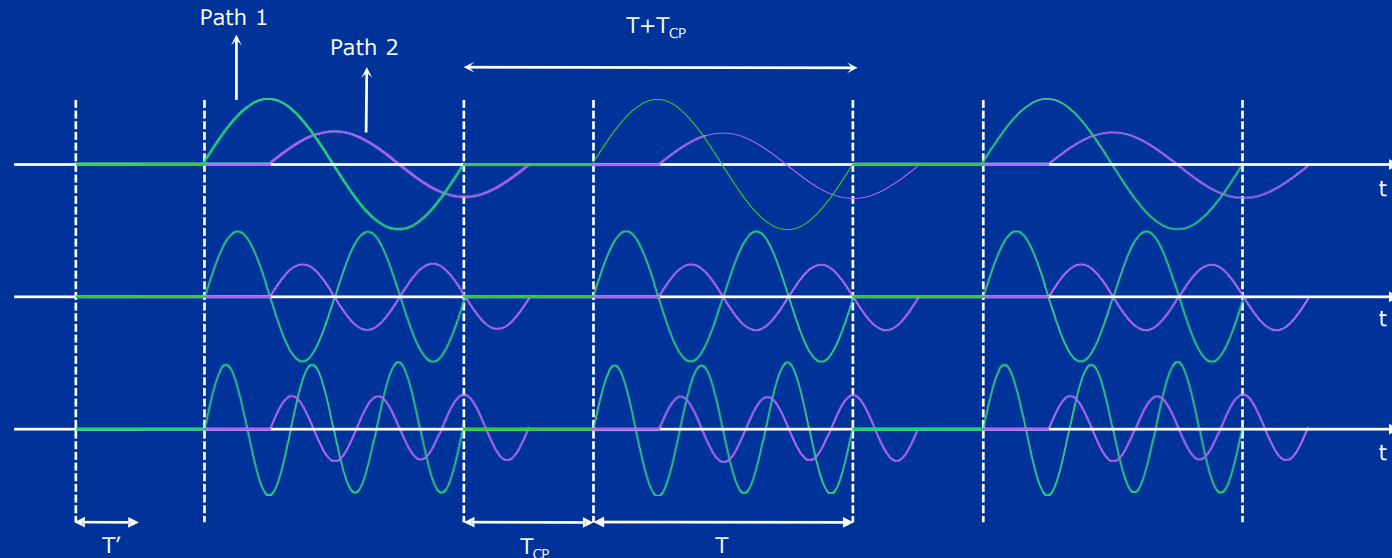
Cyclic Prefix

OFDM Signal with "Empty" Guard Time

Two-path channel
relative delay = T'



ISI is eliminated but ICI is not



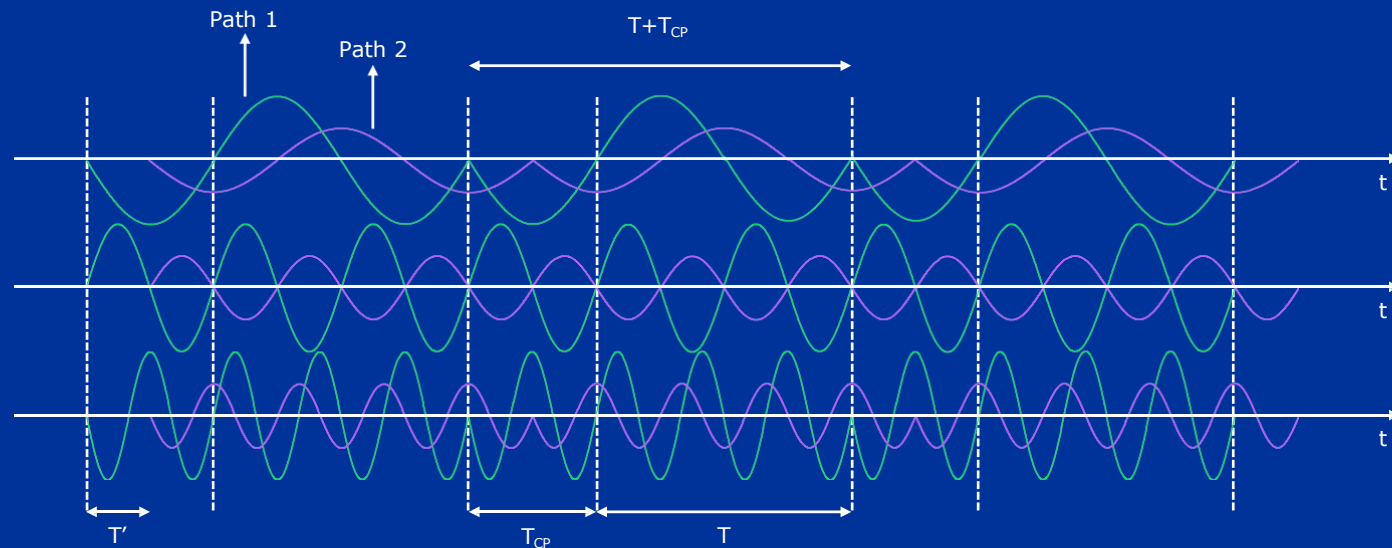
Cyclic Prefix

OFDM Signal with Cyclic Extended Guard Time

Two-path channel
relative delay = T'



Both ISI and ICI are eliminated



Cyclic Prefix

- With a cyclic prefix, the received symbol in frame k and sub-channel n , $R_{k,n}$, is related to the transmitted symbol in the same frame and sub-channel, $S_{k,n}$, by

$$R_{k,n} = H_{k,n} S_{k,n} + N_{k,n}$$

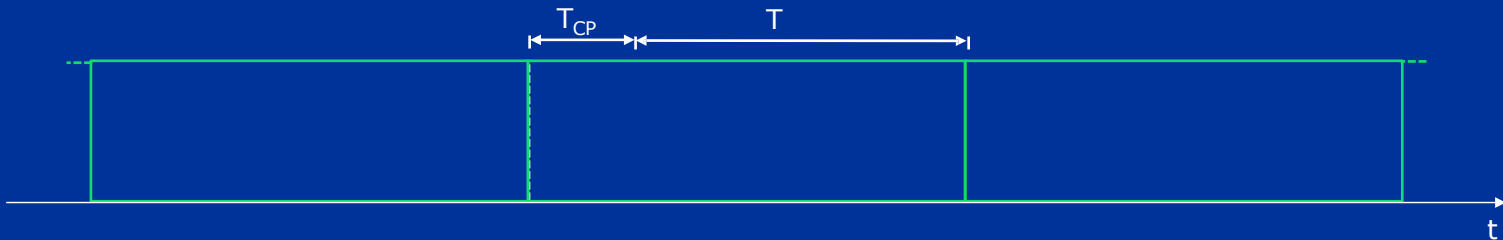
where $H_{k,n}$ is the channel frequency response in frame k and sub-channel n and $N_{k,n}$ is the noise.

- Insertion of an appropriate cyclic prefix eliminates ISI and ICI in a multipath channel but it also introduces a loss in the SNR and data rate.
- The SNR loss is equal to $10 \times \log_{10}(1 + T_{CP}/T)$ (dB)
- The data rate loss is equal to $(1 + T_{CP}/T)^{-1}$ ($\times 100\%$)

Windowing

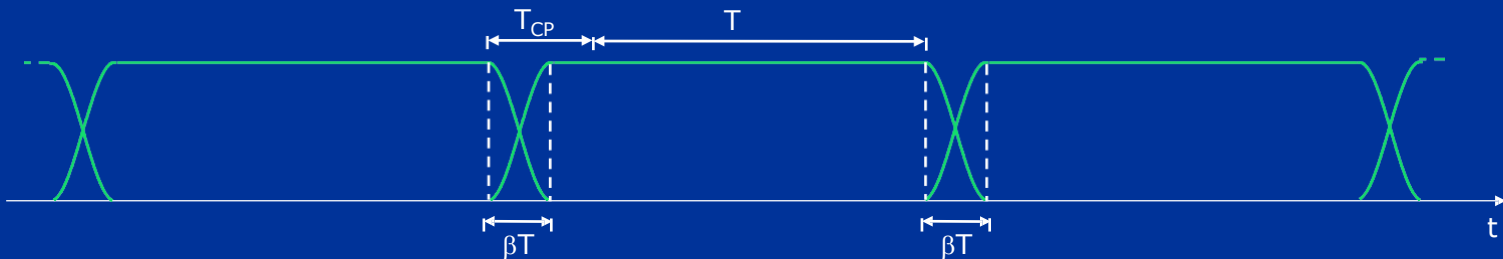
OFDM Signal with Rectangular Window

Sharp phase transitions $\Rightarrow$ High out-of-band radiation ... Delay spread tolerance = T_{CP}



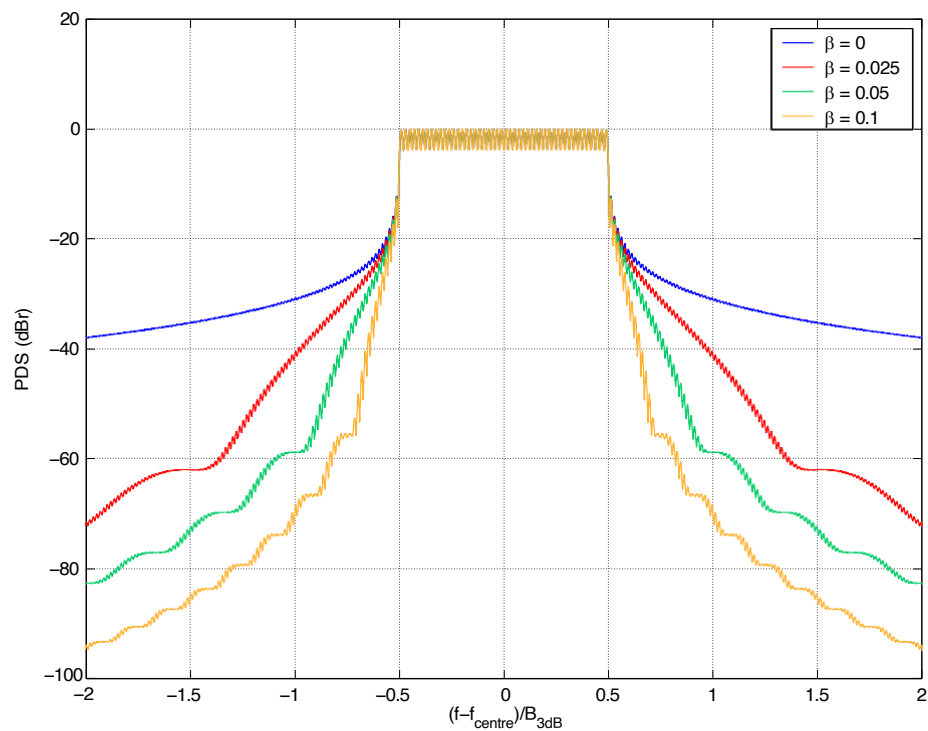
OFDM Signal with Raised Cosine Window

Smooth phase transitions $\Rightarrow$ Low out-of-band radiation ... Delay spread tolerance = $T_{CP} - \beta T$



Windowing

Power Density Spectrum of “Windowed” OFDM Signals



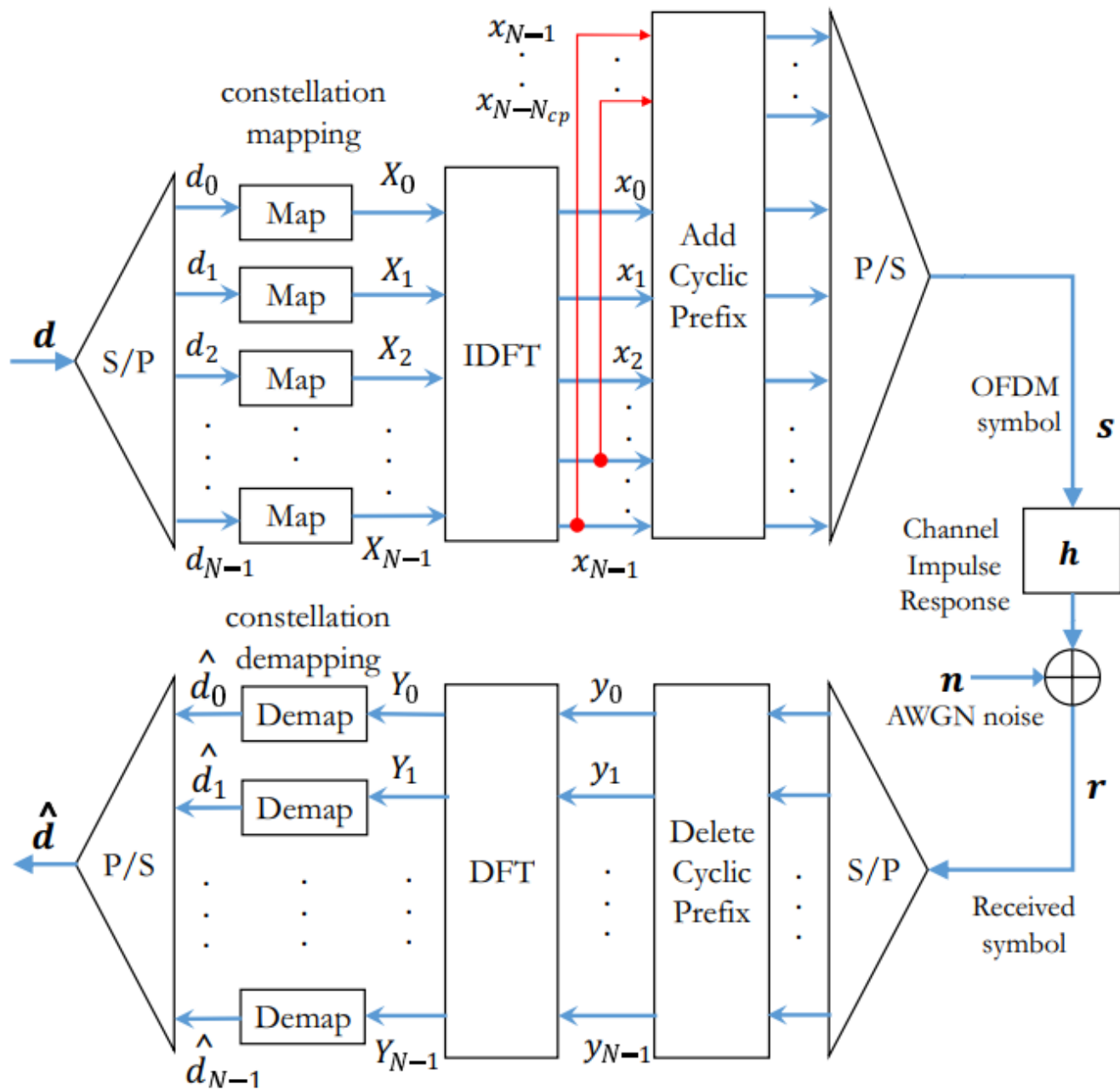


Figure: Discrete-time simulation model for OFDM transmission and reception

#1 Use the circular convolution properties of DFT

$$h[n] \circledast s[n] \equiv H[k]S[k]$$

#2 Create two randomized vectors for $s[n]$ and $h[n]$. $s[n]$ is of length 8, the channel impulse response $h[n]$ is of length 3 and we intend to use $N = 8$ -point DFT/IDFT.

#3 Convolve the vectors s and h linearly and circularly.

$$r[n] = h[n] * s[n] + w[n] \quad (* \rightarrow \text{linear convolution})$$

$$r[n] = h[n] \circledast s[n] \quad (\circledast \rightarrow \text{circular convolution})$$

using a simple frequency domain equalizer (actually this is a zero-forcing equalizer when viewed in time domain) that just inverts the *estimated* channel response and multiplies it with the frequency response of the received signal

$$\hat{S}[k] = \frac{R[k]}{H[k]}$$

The simple frequency domain equalizer is possible only if the channel performs circular convolution.

But, all channels perform linear convolution.

The linear convolution can be converted into circular convolution by adding *cyclic prefix* (CP) in the OFDM architecture.

#4 Append a cyclic prefix to the signal **s** by copying last **N_cp** symbols from **s** and pasting it to its front.

#5 Since the channel delay is 3 symbols (CIR of length 3), add atleast 2 CP symbols.

#6 Sending the cyclic-prefixed OFDM symbol through a channel that performs linear filtering.

#7 Compare the circular convolution and linear convolution with cyclic prefixed signal.

Verify the same $IDFT[DFT(h[n]) DFT(s[n])]$ is same as circular convolution of $h(n)$ and $s(n)$

$$\begin{aligned} r[n] &= IDFT \{R[k]\} \\ &= IDFT \{H[k]S[k]\} \\ &= IDFT \{DFT(h[n]) DFT(s[n])\} \\ &= h[n] \circledast s[n] \end{aligned}$$

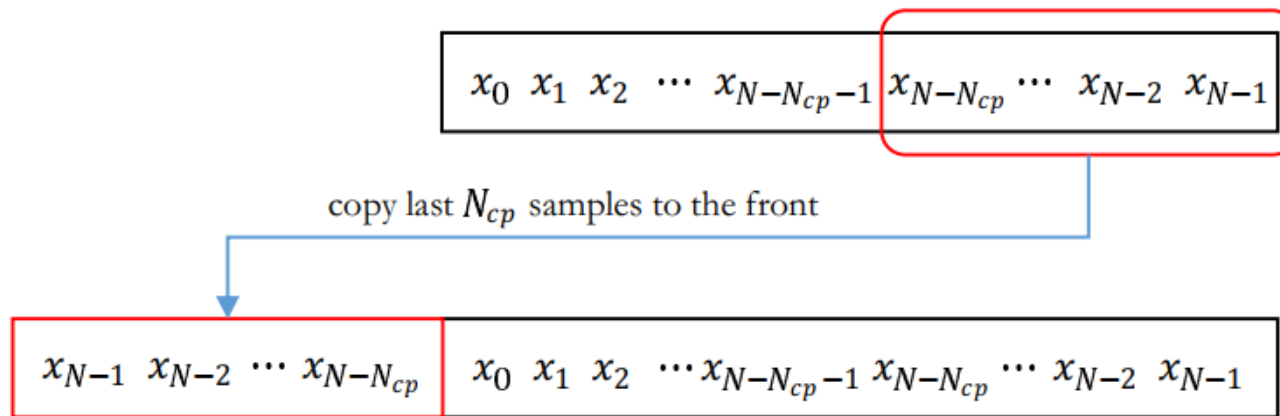


Figure: Adding a cyclic prefix in CP-OFDM

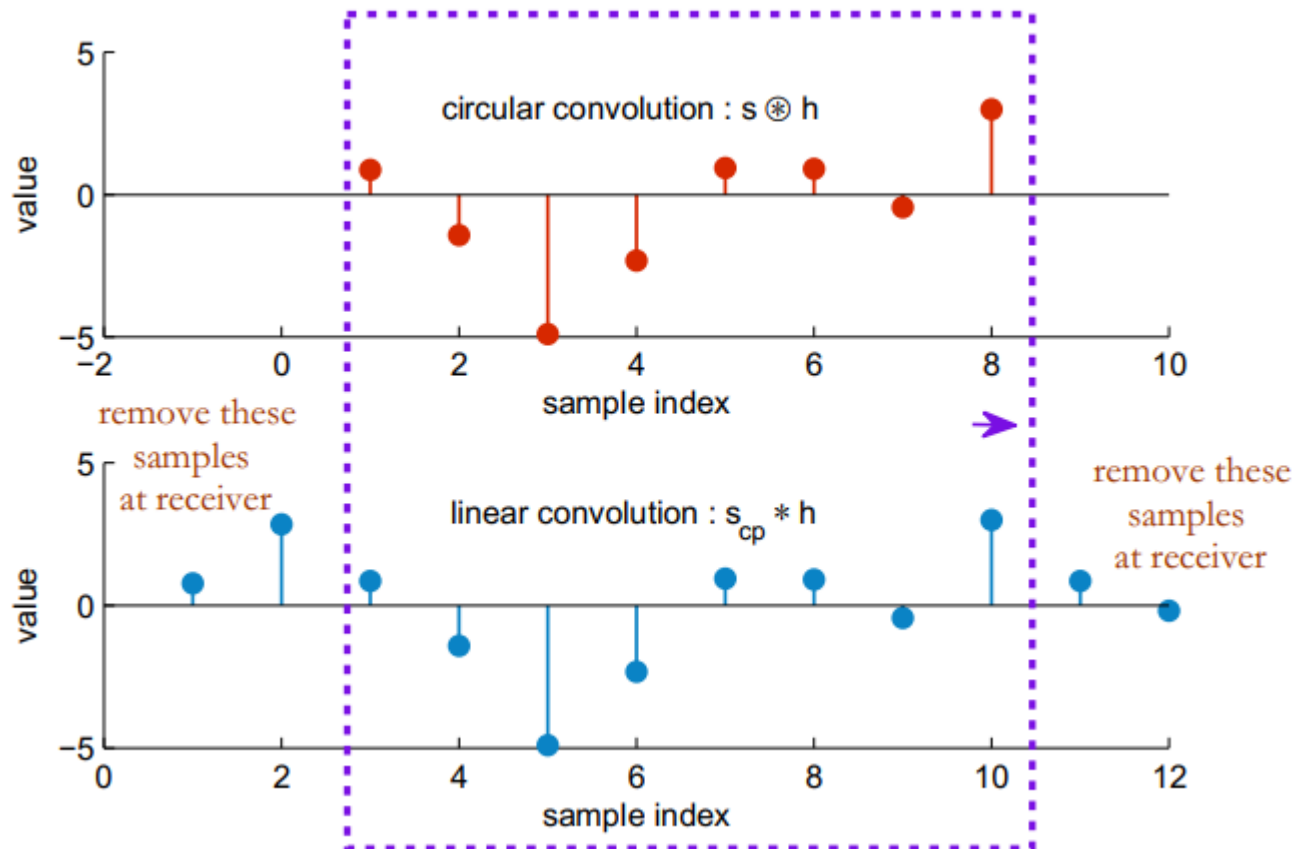


Fig: Equivalence of circular convolution and linear convolution with cyclic prefixed signal